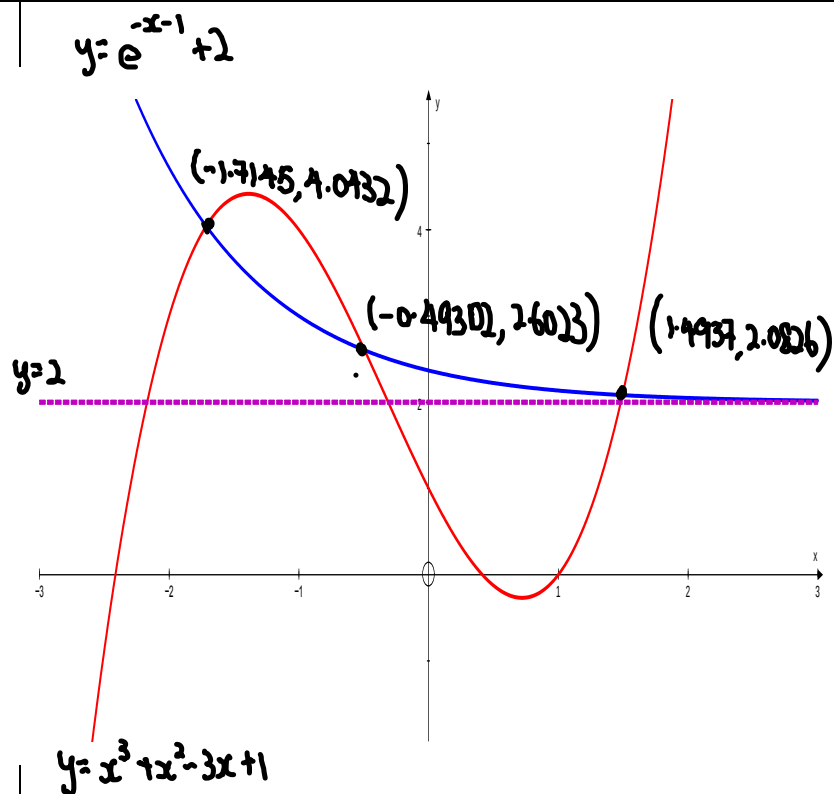


SAJC H1 Prelim 2013 Solution:

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | <p>Height of stack is 140 cm <math>\Rightarrow x + y + 20 = 140</math><br/> <math>\Rightarrow x + y = 120</math><br/> <math>\Rightarrow y = 120 - x</math> ....(1)</p> <p>Given <math>(x - y)^2 = 3(x - 20)</math> ....(2)</p> <p>Sub (1) into (2):<br/> <math>(x - (120 - x))^2 = 3x - 60</math><br/> <math>(2x - 120)^2 = 3x - 60</math><br/> <math>4x^2 - 480x + 14400 - 3x + 60 = 0</math><br/> <math>4x^2 - 483x + 14460 = 0</math><br/> Using GC, <math>x = 54.884952 \approx 54.9</math> or <math>65.865048 \approx 65.9</math> (3 s.f.)<br/> (1): <math>y = 65.1</math> or <math>54.1</math> (3 s.f.)<br/> <math>\therefore x &gt; y</math>, <math>x = 65.9</math> &amp; <math>y = 54.1</math></p> |
| 2i) | <p><math>42^2 - 4(k)(9k) &lt; 0</math> and <math>k &gt; 0</math><br/> <math>42^2 - 36k^2 &lt; 0</math><br/> <math>36k^2 - 42^2 &gt; 0</math><br/> <math>(6k - 42)(6k + 42) &gt; 0</math><br/> <math>k &lt; -7</math> or <math>k &gt; 7</math><br/> Thus,<br/> <math>k &gt; 7</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                    |
| ii) | <p><math>\frac{10x^2 + 132x}{x - 1} &lt; 90</math><br/> <math>\frac{10x^2 + 132x - 90x + 90}{x - 1} &lt; 0</math><br/> <math>\frac{10x^2 + 42x + 90}{x - 1} &lt; 0</math><br/> Since <math>10x^2 + 42x + 90</math> is always positive from (i),<br/> <math>x &lt; 1</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                             |

3i)



ii)

$$\begin{aligned}
 \text{Area enclosed} &= \int_{-1.7145}^{-0.49302} (x^3 + x^2 - 3x + 1) - (e^{-x-1} + 2) dx \\
 &\quad + \int_{-0.49302}^{1.4937} (e^{-x-1} + 2) - (x^3 + x^2 - 3x + 1) dx \\
 &= 0.8769 + 3.1080 \\
 &= 3.98 \text{ units}^2
 \end{aligned}$$

4i)

Volume,  $V = x(3x)(y) = 3x^2y$

Using Pythagoras Theorem,

$$x^2 + y^2 = (\sqrt{2})^2$$

$$y = \sqrt{2 - x^2}$$

Therefore,  $V = 3x^2\sqrt{2 - x^2}$

$$V = 3\sqrt{(2x^4 - x^6)}$$

ii)

$$V = 3(2x^4 - x^6)^{1/2}$$

$$\frac{dV}{dx} = \frac{3}{2}(2x^4 - x^6)^{-1/2}(8x^3 - 6x^5)$$

For stationary point,  $\frac{dV}{dx} = 0$

$$\frac{dV}{dx} = \frac{3}{2}(2x^4 - x^6)^{-1/2}(8x^3 - 6x^5) = 0$$

$$3x^3(4 - 3x^2) = 0$$

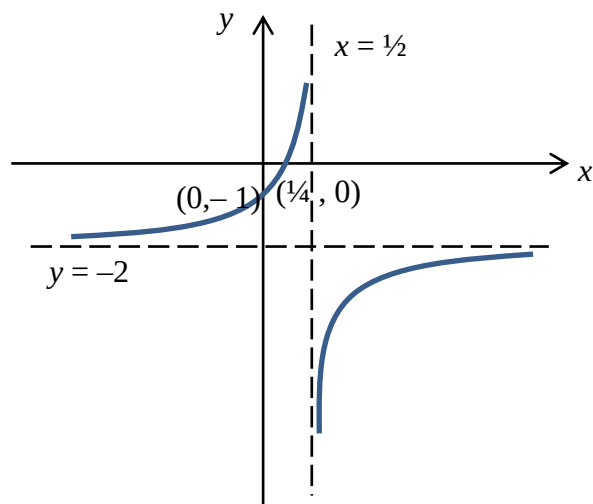
$$x = \frac{2}{\sqrt{3}} \text{ since } x > 0$$

| $x$             | $\left(\frac{2}{\sqrt{3}}\right)^-$ | $x = \frac{2}{\sqrt{3}}$ | $\left(\frac{2}{\sqrt{3}}\right)^+$ |
|-----------------|-------------------------------------|--------------------------|-------------------------------------|
| $\frac{dV}{dx}$ | +                                   | 0                        | -                                   |
| GI              | /                                   | -                        | \                                   |

When  $x = \frac{2}{\sqrt{3}}$ , the volume is maximum.

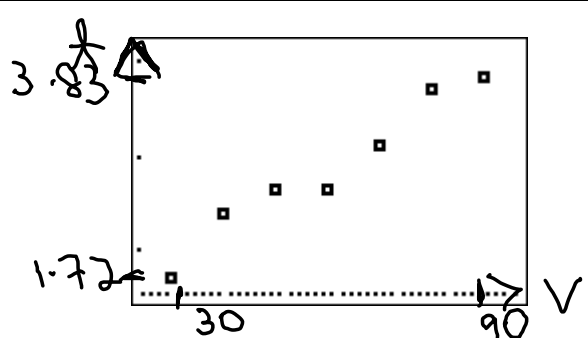
$$\text{maximum } V = 3\sqrt{\frac{32}{27}} \text{ cm}^3.$$

5(i)



|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (ii)  | <p>At A, <math>-2.5 = \frac{1-4x}{2x-1}</math><br/> <math>-5x + 2.5 = 1 - 4x</math><br/> <math>x = 1.5</math><br/> i.e. A(1.5, -2.5)<br/> At A, <math>\frac{dy}{dx} = \frac{1}{2}</math> (Using GC)</p> <p>OR</p> $y = -2 - (2x - 1)^{-1}$ $\frac{dy}{dx} = (2x - 1)^{-2} (2)$ <p>At A, <math>\frac{dy}{dx} = \frac{1}{2}</math></p>                                                                                                                                                                                                                                                                                                      |
| (iii) | <p>Equation of normal at A is<br/> <math>y = -2x + c</math><br/> <math>-2.5 = -2(1.5) + c</math><br/> <math>c = 0.5</math><br/> <math>y = -2x + 0.5</math><br/> When <math>y = 0</math>, <math>x = 0.25</math><br/> <math>\therefore P(0.25, 0)</math><br/> Equation of tangent at A is<br/> <math>y = \frac{1}{2}x + c</math><br/> <math>-2.5 = \frac{1}{2}(1.5) + c</math><br/> <math>c = -3.25</math><br/> <math>\therefore Q(0, -3.25)</math></p> <p>Area of triangle OPQ<br/> <math>= \frac{1}{2} \times 0.25 \times 3.25</math><br/> <math>= 0.40625 \text{ units}^2 \quad \text{or} \quad \frac{13}{32} \text{ units}^2</math></p> |
| 6(i)  | <p>Let A and B be the events “girl has doll” and “girl owns pet” respectively.<br/> <math>P(A) = 0.7</math>, <math>P(A B) = 0.6</math><br/> Since <math>P(A B) \neq P(A)</math>, the two events are not independent.</p>                                                                                                                                                                                                                                                                                                                                                                                                                  |
| (iia) | $P(A B) = \frac{P(A \cap B)}{P(B)}$ $0.6 = \frac{P(A \cap B)}{p}$ $P(A \cap B) = 0.6p$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| (iib) | <p>Required probability<br/> <math>= P(A) + P(B) - P(A \cap B)</math><br/> <math>= 0.7 + p - 0.6 \times p</math><br/> <math>= 0.7 + 0.4p</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|       |                                                                                                                                                                                                                                                                                                                     |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | <p>As probability cannot exceed 1,</p> $0.7 + 0.4p \leq 1$ $0.4p \leq 0.3$ $p \leq 0.75$                                                                                                                                                                                                                            |
| 7(i)  | As only the students who visited the gym get a chance of being selected, the sampling method is not random.                                                                                                                                                                                                         |
| (ii)  | <p>Use a stratified sampling. Select a sample of 10, 20, 5, 15 from each of Year 1 Arts, Year 1 Science, Year 2 Arts &amp; Year 2 Science.</p> <p>Within each stratum, pick the required number using simple random sampling.</p>                                                                                   |
| 8(i)  | <div style="text-align: center;"> <math>\underline{1^{\text{st}} \text{ attempt}} \qquad \underline{2^{\text{nd}} \text{ attempt}} \qquad \underline{3^{\text{rd}} \text{ attempt}}</math> </div>                                                                                                                   |
| (ii)  | $0.9 - p = 0.6$ $\Rightarrow p = 0.3$                                                                                                                                                                                                                                                                               |
| (iii) | $P(\text{successful}) = 0.3 + (0.7 \times 0.4) + (0.7 \times 0.6 \times 0.5)$ $= 0.79$                                                                                                                                                                                                                              |
| 9(i)  | <p>Let <math>X</math> be the r.v. “no of students who wear glasses out of 20 students”</p> $X \sim B(20, 0.45)$ $P(X > 15) = 1 - P(X \leq 15)$ $= 0.00153$                                                                                                                                                          |
| (iia) | <p>Let <math>Y</math> be the r.v. “no of students who wear glasses out of 60 students”</p> $Y \sim B(60, 0.45)$ <p>Since <math>n = 60</math>, <math>np = 27 &gt; 5</math>, <math>nq = 33 &gt; 5</math>,</p> $Y \sim N(27, 14.85) \text{ approximately}$ $P(Y > 20) \xrightarrow{\text{c.c.}} P(Y > 20.5)$ $= 0.954$ |

|       |                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (iib) | $[P(X > 5)]^2 \times P(X > 10) \times 3$ $= [1 - P(X \leq 5)]^2 \times [1 - P(X \leq 10)] \times 3$ $= 0.667$                                                                                                                                                                                                                                                                                           |
| (iv)  | The probability found in (a) is more than (b) because all the cases in (b) are included plus additional cases. For example, there could be 12 students wearing glasses in two class and none in the other class. It is one of the cases in (a) but not (b).                                                                                                                                             |
| 10(i) | $\bar{v} = \frac{30 + 40 + 50 + 60 + 70 + 80 + 90}{7} = 60$ $\bar{t} = \frac{1.72 + 2.38 + 2.66 + 2.65 + 3.12 + k + 3.83}{7}$ $= \frac{16.36 + k}{7}$ $\bar{t} = 0.84 + 0.033786\bar{v}$ $\frac{16.36 + k}{7} = 0.84 + 0.033786(60)$ $k = 3.71$                                                                                                                                                         |
| (ii)  |                                                                                                                                                                                                                                                                                                                      |
| (iii) | <div style="border: 1px solid black; padding: 5px; width: fit-content;"> <b>Linkes</b><br/> <math>y = a + bx</math><br/> <math>a = .8421428571</math><br/> <math>b = .03375</math><br/> <math>r^2 = .9535486392</math><br/> <math>r = .9764981512</math> </div> <p>Thus, <math>r = 0.976</math>.</p> <p>a) There is a positive strong linear correlation between <math>v</math> and <math>t</math>.</p> |
| (iv)  | $t = 0.84 + 0.033786v$ <p>b) When <math>v = 55</math>, <math>t = 2.70s</math></p> <p>When <math>t = 4s</math>, <math>v = 93.5 \text{ km / h}</math></p>                                                                                                                                                                                                                                                 |
| (v)   | <p>For a), the estimate is a reliable one as the value of <math>r</math> is close to 1 and <math>v = 55</math> is within the data range.</p> <p>For b), the estimate is not reliable as <math>t = 4</math> is outside the data range.</p>                                                                                                                                                               |

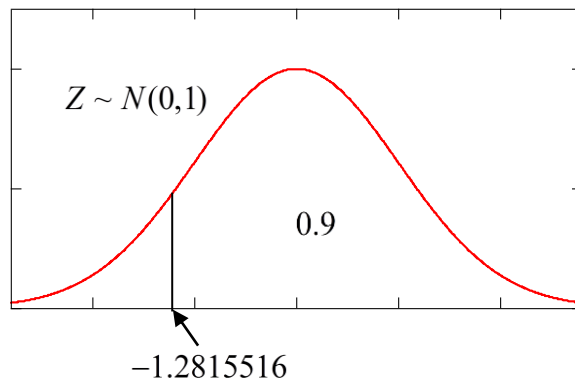
|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11(i)   | <p>Unbiased estimate of population mean = 44.11kg</p> <p>Unbiased estimate of population variance = <math>\frac{60}{59} \times 3^2 = 9.1525 \approx 9.15 \text{ kg}^2</math></p>                                                                                                                                                                                                                                                                                                                                                                                              |
| (ii)    | <p>Let <math>M</math> be the r.v. “mass of a bale of hay in kg”.</p> <p>Since <math>n = 50</math> is large,</p> $\bar{M} \sim N(44.11, \frac{9.1525}{50})$ <p>approximation by central limit theorem</p> <p><math>P(\bar{M} &gt; 44.5) = 0.181</math></p>                                                                                                                                                                                                                                                                                                                     |
| (iii)   | <p>Let <math>\mu</math> be the population mean mass.</p> <p>Test <math>H_0 : \mu = 45</math></p> <p>against <math>H_1 : \mu &lt; 45</math></p> <p>As <math>n = 60</math> is large, CLT can be applied.</p> <p><math>\therefore</math> Under <math>H_0</math>, <math>\bar{M} \sim N(45, \frac{9.1525}{60})</math> approximately</p> <p>Using a left-tailed z-test at 5% level, <math>p</math>-value = 0.0113</p> <p>As <math>p</math>-value &lt; 0.05, reject <math>H_0</math>.</p> <p>At 5% level of significance, there is sufficient evidence that his belief is valid.</p> |
| (iv)    | <p>Let the mean mass of this sample be <math>\bar{x}</math>.</p> <p>For the farmer’s belief to be accepted, <math>p</math>-value &lt; 0.05.</p> <p><math>P(\bar{M} &lt; \bar{x}) &lt; 0.05</math></p> <p><math>\bar{x} &lt; 44.36</math></p> <p>The greatest value is 44.3kg.</p>                                                                                                                                                                                                                                                                                             |
| 12(i)   | <p>Let <math>X</math> be the mass (g) of a packet of biscuits and <math>E(X) = 40 \text{ g}</math> and <math>\text{var}(X) = 2.5^2</math></p> <p><math>S = X_1 + X_2 + \dots + X_{55}</math></p> <p>Since <math>n = 55</math> is large</p> <p><math>S \sim N(2200, 343.75)</math> approximately by Central Limit Theorem</p> <p><math>P(S &gt; 2230) = 0.052823 \approx 0.0528</math> (3 sig. fig.)</p>                                                                                                                                                                       |
| 12(ii)  | <p>Let <math>Y</math> be the number of cartons out of 50, each weighing more than 2230 g</p> <p><math>Y \sim B(50, 0.052823)</math></p> <p>Required probability</p> <p><math>= P(Y &lt; 7) = P(Y \leq 6) = 0.98444 \approx 0.984</math> (3 sig. fig.)</p>                                                                                                                                                                                                                                                                                                                     |
| 12(iii) | <p><math>S_1 + S_2 + S_3 - 3S_4 \sim N(0, 3(343.75) + (3^2)(343.75))</math> approximately</p> <p>i.e. <math>S_1 + S_2 + S_3 - 3S_4 \sim N(0, 4125)</math> approximately</p> <p>Required probability</p> <p><math>= P(S_1 + S_2 + S_3 - 3S_4 &gt; 10)</math></p> <p><math>= 0.43813 \approx 0.438</math> (3 sig. fig.)</p>                                                                                                                                                                                                                                                     |

12(iv)

Let  $E(X) = \mu$  then

$S \sim N(55\mu, 343.75)$  approximately

$P(S \geq 2500) \geq 0.9$



$$P\left(Z \geq \frac{2500 - 55\mu}{\sqrt{343.75}}\right) \geq 0.9$$

$$P\left(Z < \frac{2500 - 55\mu}{\sqrt{343.75}}\right) \leq 0.1$$

Using GC,  $\frac{2500 - 55\mu}{\sqrt{343.75}} \leq -1.2815516$

$$\mu \geq 45.887$$

Least  $\mu = 46$  g